Support-Function Harmonics for AR Sliding-Window Geometry

Bounded-support theory and convex-hull empirical analogues

Working notes for the Master's thesis

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Central construction. For bounded innovations, the population sliding-window support is a zonoid. Its empirical replacement throughout these notes is the ordinary convex hull of the observed sliding-window point cloud:

$$\mathcal{Z}^k \quad \longmapsto \quad \widehat{K}_{k,T} = \operatorname{conv}\{Y_t^{(k)} : t = k, \dots, T\}, \qquad \widehat{h}_{k,T}(u) = \max_{k \le t \le T} u^\top Y_t^{(k)}.$$

Projecting this support function onto spherical harmonics gives empirical size and shape descriptors. Degree zero gives half the mean width, degree two gives quadratic anisotropy, and degree four captures finer directional shape. All empirical quantities below are functionals of this same convex hull.

1 AR setup and the bounded population support

Consider the stationary causal AR(p) process

$$X_t = \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + Z_t, \qquad Z_t \overset{\text{i.i.d.}}{\sim} U[-1, 1].$$

Assume $\Phi(z) = 1 - \sum_{r=1}^{p} \phi_r z^r$ has no zeros for $|z| \le 1$. The impulse-response coefficients satisfy

$$\psi_0 = 1, \qquad \psi_j = \sum_{r=1}^{p} \phi_r \psi_{j-r} \ (j \ge 1), \qquad \psi_j = 0 \ (j < 0), \qquad \sum_{j \ge 0} |\psi_j| < \infty.$$

For $1 \le k \le T$, put

$$Y_t^{(k)} = (X_t, X_{t-1}, \dots, X_{t-k+1})^\top, \qquad g_j^{(k)} = (\psi_j, \psi_{j-1}, \dots, \psi_{j-k+1})^\top.$$

Then $Y_t^{(k)} = \sum_{j \ge 0} g_j^{(k)} Z_{t-j}$. Independence and the full innovation support $[-1, 1]$ yield the compact, origin-symmetric support body

$$\boxed{\mathcal{Z}^k = \sum_{j=0}^{\infty} [-g_j^{(k)}, g_j^{(k)}], \qquad h_k(u) = \sum_{j=0}^{\infty} |u^\top g_j^{(k)}|.}$$

The infinite Minkowski sum is understood as its Hausdorff limit; convergence follows from $\sum_j \left\| g_j^{(k)} \right\| < \infty$. The empirical hull lies in $\mathcal{Z}^k$ almost surely. The uniform model fixes the innovation half-width at one; replacing $[-1, 1]$ by $[-b, b]$, $b > 0$, multiplies population support functions and their harmonic components by b.

2 Harmonic components and finite-sample symmetry

For $k \geq 2$, let σ_{k-1} be normalized surface measure on $\mathbb{S}^{k-1}$, with total mass one. Write $\mathcal{H}_n^{(k)}$ for the spherical harmonics of degree n, and

$$N(k,n) = \binom{k+n-1}{n} - \binom{k+n-3}{n-2}, \qquad N(k,0) = 1, \quad N(k,1) = k.$$

The second binomial term is zero for $n < 2$. For $k = 2$, $N(2,n) = 2$ for $n \geq 1$; the expansion is the usual Fourier series in the angle of u. For an orthonormal basis $\{Y_{n,m}\}_{m=1}^{N(k,n)}$, define

$$a_{n,m}(K) = \int_{\mathbb{S}^{k-1}} h_K(v) Y_{n,m}(v)\, d\sigma(v), \qquad Q_n[K](u) = \sum_{m=1}^{N(k,n)} a_{n,m}(K) Y_{n,m}(u).$$

Thus $h_K = \sum_{n\geq 0} Q_n[K]$ in $L^2(\sigma)$ and $\|Q_n[K]\|_{L^2}^2 = \sum_m |a_{n,m}(K)|^2$. Each $Q_n[K]$ is a function; its norm is a basis-independent scalar descriptor. We use

$$Q_n^{(k)} := Q_n[\mathcal{Z}^k], \qquad \widehat{Q}_n^{(k,T)} := Q_n[\widehat{K}_{k,T}].$$

One projection formula for both bodies

Let $P_n^{(k)}(1) = 1$ be the normalized zonal polynomial: a normalized Gegenbauer polynomial for $k \geq 3$, and $P_n^{(2)}(\cos\theta) = \cos(n\theta)$ for $k = 2$. The addition theorem gives

$$\boxed{Q_n[K](u) = N(k,n) \int_{\mathbb{S}^{k-1}} h_K(v) P_n^{(k)}(u^\top v)\, d\sigma(v).}$$

In particular,

$$P_2^{(k)}(t) = \frac{kt^2 - 1}{k-1}, \qquad P_4^{(k)}(t) = \frac{(k+2)(k+4)t^4 - 6(k+2)t^2 + 3}{(k-1)(k+1)}.$$

The raw hull and its even support component

The population body is origin-symmetric, so $Q_{2r+1}^{(k)} = 0$. A finite-sample hull usually has nonzero odd components. Its even part is

$$\boxed{\widehat{h}_{k,T}^{\mathrm{e}}(u) = \frac{\widehat{h}_{k,T}(u) + \widehat{h}_{k,T}(-u)}{2} = \frac{\max_t u^\top Y_t^{(k)} - \min_t u^\top Y_t^{(k)}}{2}.}$$

This is the support function of $\frac{1}{2}(\widehat{K}_{k,T} - \widehat{K}_{k,T})$. Using it in an even-degree projection leaves the raw-hull result unchanged:

$$\widehat{Q}_{2r}^{(k,T)} = \Pi_{2r} \widehat{h}_{k,T}^{\mathrm{e}}, \qquad \widehat{h}_{k,T}^{\mathrm{e}} = \widehat{Q}_0^{(k,T)} + \widehat{Q}_2^{(k,T)} + \widehat{Q}_4^{(k,T)} + \cdots \quad \text{in } L^2.$$

This is a way to compute the original hull's even harmonics and widths. It is different from replacing the sample by $\mathrm{conv}\{\pm Y_t^{(k)}\}$, whose support function is $\max_t |u^\top Y_t^{(k)}|$ and whose widths generally differ.

3 Bounded-support formulas for the theoretical target

These formulas describe $\mathcal{Z}^k$ and provide a benchmark for the empirical hull. Write $g_j^{(k)} = r_j v_j$, where $r_j = \left\|g_j^{(k)}\right\|$ and $v_j = g_j^{(k)}/r_j$; omit zero generators. The even generating measure is

$$\rho_k = \frac{1}{2}\sum_{j\geq 0} r_j(\delta_{v_j} + \delta_{-v_j}), \qquad h_k(u) = \int_{\mathbb{S}^{k-1}} |u^\top v|\, \rho_k(dv).$$

The spherical cosine transform acts diagonally on each harmonic degree. Put

$$\mu_k = \frac{\Gamma(k/2)}{\sqrt{\pi}\,\Gamma((k+1)/2)}, \qquad \lambda_{n,k} = \int_{\mathbb{S}^{k-1}} |v_1| P_n^{(k)}(v_1)\, d\sigma(v).$$

Then

$$\lambda_{0,k} = \mu_k, \qquad \lambda_{2,k} = \frac{\mu_k}{k+1}, \qquad \lambda_{4,k} = -\frac{\mu_k}{(k+1)(k+3)},$$

and $Q_n^{(k)}(u) = N(k,n)\lambda_{n,k}\sum_j r_j P_n^{(k)}(u^\top v_j)$.

Degree zero: mean width

With $S_k = \sum_j r_j$,

$$\boxed{Q_0^{(k)} = \mu_k S_k, \qquad \overline{w}(\mathcal{Z}^k) = 2Q_0^{(k)} = 2\mu_k \sum_j \left\|g_j^{(k)}\right\|.}$$

Equivalently, $\mu_k = 2\kappa_{k-1}/(k\kappa_k)$, where κ_k is the volume of the unit ball in $\mathbb{R}^k$. Since $V_1(\mathcal{Z}^k) = 2\sum_j r_j$, the mean width equals $\mu_k V_1(\mathcal{Z}^k)$.

Degree two: quadratic anisotropy

Define

$$M_k = \sum_{j\geq 0} \frac{g_j^{(k)} g_j^{(k)\top}}{\left\|g_j^{(k)}\right\|}, \qquad A_k = M_k - \frac{S_k}{k} I_k, \qquad c_k = \frac{\mu_k k(k+2)}{2(k+1)}.$$

Then $\operatorname{tr} M_k = S_k$, $\operatorname{tr} A_k = 0$, and

$$\boxed{Q_2^{(k)}(u) = c_k u^\top A_k u, \qquad \left\|Q_2^{(k)}\right\|_{L^2}^2 = \mu_k^2 \frac{k(k+2)}{2(k+1)^2}\, \|A_k\|_{\mathrm{F}}^2.}$$

The matrix $C_k := c_k A_k$ is the direct quadratic harmonic coefficient matrix.

Degree four: finer directional shape

With $N(k,4) = k(k-1)(k+1)(k+6)/24$,

$$Q_4^{(k)}(u) = N(k,4)\lambda_{4,k}\sum_j r_j P_4^{(k)}(u^\top v_j).$$

For any degree n, the corresponding generator identity is

$$\left\|Q_n^{(k)}\right\|_{L^2}^2 = N(k,n)\lambda_{n,k}^2 \sum_{i,j\geq 0} r_i r_j P_n^{(k)}(v_i^\top v_j).$$

These generator formulas are specific to the population support representation. The empirical analogues below instead insert the hull support function into the general harmonic projection formula.

4 Direct empirical analogues of Q_0 and Q_2

For readability, suppress (k,T) within this section and write $\widehat{h} = h_{\widehat{K}_{k,T}}$, $\widehat{h}^{\mathrm{e}} = (\widehat{h}(\cdot) + \widehat{h}(-\cdot))/2$. All displayed integrals are exact functionals of the observed hull.

Empirical mean width

The degree-zero component is constant and satisfies

$$\boxed{\widehat{Q}_0 = \int_{\mathbb{S}^{k-1}} \widehat{h}(u)\, d\sigma(u) = \int_{\mathbb{S}^{k-1}} \widehat{h}^{\mathrm{e}}(u)\, d\sigma(u), \qquad \widehat{\overline{w}}_{k,T} = 2\widehat{Q}_0.}$$

Consequently,

$$\widehat{\overline{w}}_{k,T} = \int_{\mathbb{S}^{k-1}} \left(\max_t u^\top Y_t^{(k)} - \min_t u^\top Y_t^{(k)} \right) d\sigma(u).$$

This is the spherical average of the observed projection range. It depends only on the convex hull, so adding points in its interior does not change the result.

The traceless quadratic coefficient matrix

Define

$$\boxed{\widehat{C}_{k,T} = \frac{k(k+2)}{2} \int_{\mathbb{S}^{k-1}} \widehat{h}^{\mathrm{e}}(u) \left(uu^\top - \frac{1}{k} I_k \right) d\sigma(u).}$$

The same matrix results if $\widehat{h}^{\mathrm{e}}$ is replaced by $\widehat{h}$. Using the normalized spherical fourth moments gives

$$\boxed{\widehat{Q}_2^{(k,T)}(u) = u^\top \widehat{C}_{k,T} u, \qquad \left\| \widehat{Q}_2^{(k,T)} \right\|_{L^2}^2 = \frac{2}{k(k+2)} \left\| \widehat{C}_{k,T} \right\|_{\mathrm{F}}^2.}$$

The matrix is symmetric and traceless. Its eigenvectors describe the principal directions of the degree-two component, while its eigenvalues quantify signed deviations from the degree-zero baseline. In particular, $\int \widehat{Q}_2^{(k,T)}\, d\sigma = 0$.

Recovering an empirical counterpart of the support matrix M_k

To connect the hull harmonics to the population matrix in Section 3, set

$$\widehat{A}_{k,T}^{\mathrm{sup}} = \frac{\widehat{C}_{k,T}}{c_k}, \qquad \boxed{\widehat{M}_{k,T}^{\mathrm{sup}} = \frac{\widehat{C}_{k,T}}{c_k} + \frac{\widehat{Q}_0^{(k,T)}}{k\mu_k} I_k.}$$

Equivalently, if $J_{k,T} = \int \widehat{h}^{\mathrm{e}}(u) uu^\top\, d\sigma(u)$, then

$$\boxed{\widehat{M}_{k,T}^{\mathrm{sup}} = \frac{1}{\mu_k} \left((k+1) J_{k,T} - \widehat{Q}_0^{(k,T)} I_k \right), \qquad \operatorname{tr} \widehat{M}_{k,T}^{\mathrm{sup}} = \frac{\widehat{Q}_0^{(k,T)}}{\mu_k}.}$$

This is a linear reconstruction from the hull's degree-zero and degree-two harmonics. A general empirical hull need not be a zonoid, so the reconstruction does not supply a generator representation of the entire hull. It targets M_k when the empirical hull converges to $\mathcal{Z}^k$.

5 Empirical Q_4 and numerical spherical integration

The direct degree-four analogue is

$$\boxed{\widehat{Q}_4^{(k,T)}(u) = N(k,4) \int_{\mathbb{S}^{k-1}} \widehat{h}^{\mathrm{e}}_{k,T}(v) P_4^{(k)}(u^\top v)\, d\sigma(v).}$$

The addition theorem also gives the exact, basis-free energy identity

$$\boxed{\left\|\widehat{Q}_4^{(k,T)}\right\|_{L^2}^2 = N(k,4) \iint \widehat{h}^{\mathrm{e}}_{k,T}(v)\widehat{h}^{\mathrm{e}}_{k,T}(w) P_4^{(k)}(v^\top w)\, d\sigma(v) d\sigma(w).}$$

Both integrations are over the unit sphere. These expressions use directional maxima and minima of the point cloud; population generator multipliers are not inserted into this general projection formula.

A shared set of integration directions

Choose L unit directions U_ℓ with nonnegative weights ω_ℓ summing to one. Independent uniform directions with $\omega_\ell = 1/L$ provide a Monte Carlo rule. In dimension two, equally spaced angles give a simple deterministic rule. For higher dimensions, spherical designs or other refined cubature rules can be used. Evaluate

$$H_\ell = \frac{1}{2}\left(\max_t U_\ell^\top Y_t^{(k)} - \min_t U_\ell^\top Y_t^{(k)}\right), \qquad \ell = 1, \dots, L.$$

Then the numerical approximations are

$$\begin{aligned}
\widehat{Q}_0^{(k,T,L)} &= \sum_{\ell=1}^{L} \omega_\ell H_\ell, \\
\widehat{C}_{k,T}^{(L)} &= \frac{k(k+2)}{2} \sum_{\ell=1}^{L} \omega_\ell H_\ell \left(U_\ell U_\ell^\top - \frac{1}{k} I_k\right), \\
\widehat{Q}_4^{(k,T,L)}(u) &= N(k,4) \sum_{\ell=1}^{L} \omega_\ell H_\ell P_4^{(k)}(u^\top U_\ell).
\end{aligned}$$

Once these harmonic approximations have been formed, their spherical norms satisfy

$$\begin{aligned}
\left\|\widehat{Q}_2^{(k,T,L)}\right\|_{L^2}^2 &= \frac{2}{k(k+2)} \left\|\widehat{C}_{k,T}^{(L)}\right\|_{\mathrm{F}}^2, \\
\left\|\widehat{Q}_4^{(k,T,L)}\right\|_{L^2}^2 &= N(k,4) \sum_{\ell,m=1}^{L} \omega_\ell \omega_m H_\ell H_m P_4^{(k)}(U_\ell^\top U_m).
\end{aligned}$$

The last expression is nonnegative in exact arithmetic and is the exact norm of the numerically constructed degree-four function. At finite L it is an approximation to the exact hull energy; under Monte Carlo integration it is generally not an unbiased estimator of that energy.

Accuracy and cost. Support evaluation requires only projection maxima and minima, so constructing hull facets is unnecessary. Batched projections cost $O(L(T-k+1)k)$; the direct degree-four pair sum costs $O(L^2 k)$. Because the hull support function is piecewise linear on directions, a low-degree spherical design does not make these integrals exact. Increase L and compare independent direction sets, or refine the cubature, until the descriptors stabilize.

6 Minimum width, location, and shape summaries

Exact empirical width and its degree-two approximation

For the raw hull, width in direction u and minimum width are

$$\widehat{w}_{k,T}(u) = \widehat{h}_{k,T}(u) + \widehat{h}_{k,T}(-u) = 2\widehat{h}^{\mathrm{e}}_{k,T}(u), \qquad \widehat{w}_{\min,k,T} = 2 \min_{\|u\|=1} \widehat{h}^{\mathrm{e}}_{k,T}(u).$$

Keeping degrees zero and two gives

$$\boxed{\widehat{w}^{(2)}_{\min,k,T} = 2\left(\widehat{Q}_0^{(k,T)} + \lambda_{\min}(\widehat{C}_{k,T})\right).}$$

Thus $2\widehat{Q}_0$ is the mean-width baseline, and $2\lambda_{\min}(\widehat{C})$ is the leading correction toward the thinnest direction. The population counterpart is $w^{(2)}_{\min}(k) = 2(Q_0^{(k)} + c_k\lambda_{\min}(A_k))$. For the continuous residual

$$\widehat{R}_{\geq 4}(u) = \widehat{h}^{\mathrm{e}}_{k,T}(u) - \widehat{Q}_0^{(k,T)} - u^\top \widehat{C}_{k,T} u,$$

the elementary inequality $|\min f - \min g| \leq \|f - g\|_\infty$ yields

$$\boxed{|\widehat{w}_{\min,k,T} - \widehat{w}^{(2)}_{\min,k,T}| \leq 2\left\|\widehat{R}_{\geq 4}\right\|_\infty.}$$

The truncated function need not be a support function, and its minimum can be negative. A small L^2 energy in degree four alone does not bound the full uniform residual. Comparing the mean-width, degree-two minimum-width, and full minimum-width curves tests how much of a transition is already explained by Q_2.

Directional search. The smallest width among finitely many sampled directions is an upper bound on the exact empirical minimum width. It should be reported as an approximation unless the minimization has been certified. If the cloud has affine dimension below k, its exact minimum width is zero; in particular, this occurs whenever $T - k + 1 \leq k$.

Steiner point and odd harmonics

The raw hull has the empirical Steiner point

$$\widehat{s}_{k,T} = k \int_{\mathbb{S}^{k-1}} u\, \widehat{h}_{k,T}(u)\, d\sigma(u), \qquad \widehat{Q}_1^{(k,T)}(u) = u^\top \widehat{s}_{k,T}.$$

It belongs to the hull and is translation and rotation equivariant. It need not vanish in a finite sample, whereas $s(\mathcal{Z}^k) = 0$. Translating the hull by $-\widehat{s}_{k,T}$ removes degree one; it does not generally remove higher odd degrees. Even harmonics and widths are translation invariant. Their computation from $\widehat{h}^{\mathrm{e}}_{k,T}$ automatically removes all odd contributions.

Scale-free empirical shape diagnostics

When $\widehat{Q}_0^{(k,T)} > 0$, define

$$\widehat{R}_2(k) = \frac{\left\|\widehat{Q}_2^{(k,T)}\right\|_{L^2}}{\widehat{Q}_0^{(k,T)}}, \qquad \widehat{R}_4(k) = \frac{\left\|\widehat{Q}_4^{(k,T)}\right\|_{L^2}}{\widehat{Q}_0^{(k,T)}}, \qquad \widehat{D}_2(k) = -\frac{\lambda_{\min}(\widehat{C}_{k,T})}{\widehat{Q}_0^{(k,T)}}.$$

They summarize relative anisotropy. In particular, $\widehat{w}^{(2)}_{\min,k,T}/\widehat{\overline{w}}_{k,T} = 1 - \widehat{D}_2(k)$.

7 Exact AR-filter structure in the bounded-support setting

Assume $p \geq 1$ is the minimal AR order, so $\phi_p \neq 0$. At dimension $p+1$ the complete recursion first fits inside one sliding window. With $a = (1, -\phi_1, \ldots, -\phi_p)^\top$ and $u_a = a/\|a\|$,

$$a^\top Y_t^{(p+1)} = Z_t, \qquad \boxed{h_{p+1}(u_a) = \frac{1}{\|a\|}, \qquad w_{\mathcal{Z}^{p+1}}(u_a) = \frac{2}{\|a\|}.}$$

For the observed raw hull, the corresponding exact identities are

$$\widehat{h}_{p+1,T}(u_a) = \frac{\max_{t=p+1,\ldots,T} Z_t}{\|a\|}, \qquad \boxed{\widehat{w}_{p+1,T}(u_a) = \frac{\max_{t=p+1,\ldots,T} Z_t - \min_{t=p+1,\ldots,T} Z_t}{\|a\|}.}$$

This bounds the empirical minimum width from above. It does not imply that u_a is the globally thinnest direction or an eigenvector of the degree-two matrix.

A growing family of filter directions

For $k \geq p+1$, let $B_{p,k} \in \mathbb{R}^{(k-p)\times k}$ have shifted copies of $a^\top$ as its rows. Then

$$B_{p,k} Y_t^{(k)} = (Z_t, Z_{t-1}, \ldots, Z_{t-k+p+1})^\top.$$

Its row space $\mathcal{S}_{p,k} = \mathrm{range}(B_{p,k}^\top)$ has dimension $k-p$. For $c \in \mathbb{R}^{k-p} \setminus \{0\}$ and $u = B_{p,k}^\top c / \left\|B_{p,k}^\top c\right\|$, independence and the cube support of the innovation block give

$$h_k(u) = \frac{\|c\|_1}{\left\|B_{p,k}^\top c\right\|}.$$

The appearance of $1, 2, 3, \ldots$ filter directions at dimensions $p+1, p+2, p+3, \ldots$ is exact algebra. A separated eigenvalue block or a width elbow requires additional evidence.

Congruence identity for the population support matrix

The impulse-response recursion gives

$$B_{p,k} g_j^{(k)} = \begin{cases} e_{j+1}, & 0 \leq j \leq k-p-1, \\ 0, & j \geq k-p, \end{cases}$$

where $e_1, \ldots, e_{k-p}$ are standard basis vectors. Substitution into the generator formula for M_k yields

$$\boxed{B_{p,k} M_k B_{p,k}^\top = \mathrm{diag}\left(\frac{1}{\left\|g_0^{(k)}\right\|}, \ldots, \frac{1}{\left\|g_{k-p-1}^{(k)}\right\|} \right).}$$

This is a congruence identity; the filter matrix is not generally orthogonal. It does not imply repeated eigenvalues. Replacing M_k by $\widehat{M}_{k,T}^{\mathrm{sup}}$ gives an empirical diagnostic whose off-diagonal entries need not vanish at finite T.

8 Exploratory order selection from the empirical hull

Primary candidate: approximate filter congruence

For a candidate order q, let $\beta \in \mathbb{R}^q$ be its coefficient vector. Form $B_{q,k}(\beta)$ from shifted copies of $(1, -\beta_1, \ldots, -\beta_q)$. Define

$$\widehat{D}_{q,k,T}(\beta) = B_{q,k}(\beta)\widehat{M}^{\sup}_{k,T}B_{q,k}(\beta)^\top,$$

and, on a specified admissible coefficient set Θ_q,

$$\boxed{\widehat{R}^*_{q,k,T} = \inf_{\substack{\beta\in\Theta_q \\ \left\|\widehat{D}_{q,k,T}(\beta)\right\|_{\mathrm{F}}>0}} \frac{\left\|\mathrm{off}(\widehat{D}_{q,k,T}(\beta))\right\|^2_{\mathrm{F}}}{\left\|\widehat{D}_{q,k,T}(\beta)\right\|^2_{\mathrm{F}}}.}$$

Here off sets the diagonal to zero. A vanishing denominator is excluded and must be flagged if encountered numerically. For $q = 0$, take $B_{0,k} = I_k$ with no fitted coefficients. If orders $0, \ldots, q_{\max}$ are compared at one dimension, use $k \geq q_{\max} + 2$: a one-row filter gives a 1×1 matrix and therefore an automatically zero off-diagonal score.

The population score $R^*_{q,k}$ is obtained by replacing $\widehat{M}^{\sup}_{k,T}$ with M_k. At $q = p$ it is zero when Θ_p contains the true coefficients. Higher orders also attain zero when their admissible sets contain the zero-padded true filter. Thus a candidate rule seeks the *smallest* order compatible with a near-zero score, or a substantial last decrease in

$$\widehat{\Delta}_q(k) = \widehat{R}^*_{q-1,k,T} - \widehat{R}^*_{q,k,T}.$$

Identifiability against $q < p$, threshold calibration, and reliability of numerical optimization remain to be established. The exact population identity alone does not establish an order estimator. Use the same sample windows and integration directions across all candidate orders at a given k.

Secondary diagnostics: spectra and width curves

Let $\widehat{\gamma}^{(k,T)}_1 \leq \cdots \leq \widehat{\gamma}^{(k,T)}_k$ be the eigenvalues of $\widehat{C}_{k,T}$. For $1 \leq r < k$ and $\left\|\widehat{C}_{k,T}\right\|_{\mathrm{F}} > 0$, an exploratory gap statistic is

$$\widehat{G}_k(r) = \frac{\widehat{\gamma}^{(k,T)}_{k-r+1} - \widehat{\gamma}^{(k,T)}_{k-r}}{\left\|\widehat{C}_{k,T}\right\|_{\mathrm{F}}}.$$

Compare these gaps across k, and in simulations compare eigenvectors or eigenspaces with the known filter directions. A low-eigenvalue subspace of dimension $k - p$ is a hypothesis, not a consequence of the congruence identity.

The hull mean width $2\widehat{Q}^{(k,T)}_0$, the degree-two minimum-width approximation, and the full minimum width provide complementary curves. An elbow or L-method may be applied as an exploratory comparison, but no decreasing or flattening pattern is assumed. In particular, a breakpoint at $k = p + 1$ must be tested for these hull quantities. Using a different width curve does not by itself resolve the original breakpoint-selection problem.

9 Consistency and the two sources of approximation error

Fix k. For a stationary ergodic window sequence whose marginal support is the compact body $\mathcal{Z}^k$, the observed points are almost surely dense in that support as $T \to \infty$. Indeed, every open neighborhood of a support point has positive probability and is visited by ergodicity. A finite cover of the compact support then gives

$$\delta_{k,T} := d_H(\widehat{K}_{k,T}, \mathcal{Z}^k) \longrightarrow 0 \quad \text{almost surely.}$$

This applies to the stationary causal AR model in Section 1; the overlapping windows are not assumed independent. It is a fixed-dimension statement and supplies no rate uniform in growing k.

Geometric error from the finite point cloud

For compact convex bodies, $d_H(K, L) = \|h_K - h_L\|_\infty$. Orthogonal projection is contractive in $L^2(\sigma)$, hence, for every fixed harmonic degree n,

$$\boxed{\left\|\widehat{Q}_n^{(k,T)} - Q_n^{(k)}\right\|_{L^2} \le \delta_{k,T}, \qquad \left|\left\|\widehat{Q}_n^{(k,T)}\right\|_{L^2} - \left\|Q_n^{(k)}\right\|_{L^2}\right| \le \delta_{k,T}.}$$

In particular,

$$|\widehat{Q}_0^{(k,T)} - Q_0^{(k)}| \le \delta_{k,T}, \qquad \left\|\widehat{C}_{k,T} - C_k\right\|_{\mathrm{F}} \le \sqrt{\frac{k(k+2)}{2}}\,\delta_{k,T}.$$

The matrix reconstruction in Section 4 therefore converges to M_k. The exact widths satisfy

$$|\overline{w}(\widehat{K}_{k,T}) - \overline{w}(\mathcal{Z}^k)| \le 2\delta_{k,T}, \qquad |\widehat{w}_{\min,k,T} - w_{\min}(\mathcal{Z}^k)| \le 2\delta_{k,T}.$$

Since both bodies are measured in the same coordinates, these are direct comparisons. The degree-two width approximation is also consistent for its population counterpart by continuity of the smallest eigenvalue. Its truncation error relative to the full population minimum width is a separate issue.

Numerical error from finitely many directions

At fixed T, independent uniform integration directions give convergence of the numerical coefficients as $L \to \infty$. The practical error decomposes as

$$\left\|\widehat{Q}_n^{(k,T,L)} - Q_n^{(k)}\right\|_{L^2} \le \underbrace{\left\|\widehat{Q}_n^{(k,T,L)} - \widehat{Q}_n^{(k,T)}\right\|_{L^2}}_{\text{spherical integration error}} + \underbrace{\delta_{k,T}}_{\text{finite-cloud error}}.$$

Increasing L improves integration accuracy; increasing T improves coverage of the support. These should be varied separately in numerical experiments.

For the congruence criterion, compact admissible coefficient sets and denominators uniformly bounded away from zero allow matrix convergence to pass to uniform convergence of the objective and convergence of the infimum values. Consistency of selected AR orders additionally requires identifiability and a justified selection rule. Eigenspace convergence likewise requires a separating population eigengap; matrix convergence alone is insufficient.

10 Focused workflow and interpretation

1. **Fix the bounded AR model and coordinate scale.** Use a stationary sample, or document simulation burn-in. Form the raw windows $Y_t^{(k)}$ for each chosen k. Keep the coordinate order and any preprocessing explicit.
2. **Compute hull support values.** For each integration direction, record the maximum and minimum projected window. Their half-range is H_ℓ.
3. **Obtain the harmonic descriptors.** Compute $\widehat{Q}_0$, $\widehat{C}$, $\left\|\widehat{Q}_2\right\|_{L^2}$, $\left\|\widehat{Q}_4\right\|_{L^2}$, and $\widehat{M}^{\mathrm{sup}}$ with the formulas in Sections 4–5.
4. **Compare width and shape.** Plot mean width, the degree-two minimum-width approximation, a clearly labeled numerical full-width estimate, and the scale-free ratios. Check for affine degeneracy before interpreting minimum width.
5. **Test the AR-specific criterion.** Evaluate the congruence scores for candidate orders at dimensions with at least two filter rows. In simulations, compare with the exact population generator calculation. Treat spectral gaps and elbows as secondary diagnostics.
6. **Separate validation tasks.** Refine the directional integration at fixed data, then vary sample size and repeat simulations across stable coefficient regimes. Establish lower-order identifiability before claiming consistent order recovery.

For $k = 1$, the hull is the interval $[\min_t X_t, \max_t X_t]$. Its mean and minimum widths both equal the observed range, and $\widehat{Q}_0$ equals half that range. The spherical degree-two and degree-four formulas in these notes apply only to $k \geq 2$.

Object	Role in the empirical hull analysis
$\widehat{K}_{k,T}$	Convex hull of the observed sliding-window point cloud.
$\widehat{h}_{k,T}$	Directional maximum; determines the whole hull.
$\widehat{h}^{\mathrm{e}}_{k,T}$	Half the directional range; preserves all even hull harmonics.
$\widehat{Q}_0^{(k,T)}$	Half the empirical mean width.
$\widehat{C}_{k,T}$	Traceless matrix encoding the empirical degree-two component.
$\widehat{w}^{(2)}_{\min,k,T}$	Mean-width baseline plus the leading thin-direction correction.
$\widehat{Q}_4^{(k,T)}$	Degree-four directional shape from the same hull.
$\widehat{M}^{\mathrm{sup}}_{k,T}$	Reconstruction of the population support matrix from $\widehat{Q}_0 + \widehat{Q}_2$.
$\widehat{R}^*_{q,k,T}$	Proposed AR-order diagnostic based on approximate filter congruence.

References and normalization

H. Groemer, *Geometric Applications of Fourier Series and Spherical Harmonics*, Cambridge University Press, 1996. Background on the addition theorem, Funk–Hecke formula, and support-function harmonics. Publisher reference.

R. Schneider, *Integral Geometric Tools for Stochastic Geometry*, lecture notes. Background on support functions and the Steiner point. Author's lecture notes.

All spherical integrals and L^2 norms use $\sigma_{k-1}(\mathbb{S}^{k-1}) = 1$. The AR-filter identities are derived above; order-selection proposals are stated as research questions, rather than attributed to these background references.